

$$\begin{array}{ccc}
 G_1 & \xrightarrow{f} & G_2 \\
 \pi \downarrow & \circlearrowleft & \uparrow i \\
 G_1 / \ker(f) & \xrightarrow[\bar{f}]{\sim} & \operatorname{im}(f)
 \end{array}$$